

Raffaele Mineo

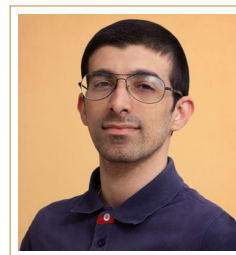
Postdoctoral Researcher in Artificial Intelligence

Department of Electrical, Electronic and Computer Engineering (DIEEI), University of Catania
Catania, Italy

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Semantic Scholar | DBLP | OpenReview | GitHub | PeRCEiVe.AI | ResearchGate | LinkedIn |
Academia.edu

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Profile

I develop artificial intelligence methods that connect perception with clinically and operationally meaningful information. My research spans cardiovascular imaging and virtual physiology, multimodal and radar-based sign language understanding, continual and federated learning, vision-language adaptation, and efficient AI for resource-constrained hardware. The common thread is the design of models that remain useful under realistic acquisition, privacy, data and compute constraints, with attention to translational and deployable outcomes.

Academic positions and research experience

2024–present

Postdoctoral Researcher / Research Fellow

University of Catania — DIEEI

HORIZON Europe NEUROKIT2E: knowledge distillation and meta-learning for on-chip AI inference; medical imaging, multimodal learning, efficient/embedded AI and project research activities.

2019–present

Research member / assistant

PeRCEiVe.AI Lab, University of Catania

Computer vision, pattern recognition, medical imaging, multimodal AI and applied deep learning.

2023–present

AI consulting and industrial R&D collaboration

DeepSensing S.r.l., Catania

Deep-learning systems for industrial inspection; multi-GPU experimentation; edge and neuromorphic computing; predictive reliability; Horizon-project activities and KPI-oriented validation.

from 2023

Research collaboration

STMicroelectronics R&D Power and Discrete, Catania

Artificial intelligence for modeling and predictive reliability in automotive and power-electronics contexts.

Education

**completed
2026**

National PhD in Artificial Intelligence — Health and Life Sciences

University Campus Bio-Medico of Rome, with University of Catania affiliation

37th cycle.

2019–2021

M.Sc. in Computer Engineering

University of Catania

110/110 cum laude. Thesis: *Prediction of paroxysmal events through an anomaly detection approach based on Transformers.*

2016–2019

B.Sc. in Computer Engineering

University of Catania

99/110. Thesis on the design and implementation of an optical data-communication interface for Li-Fi applications.

Research themes

- **Cardiovascular AI and virtual physiology:** coronary angiography, FFR/iFR estimation, spatio-temporal modeling, clinically meaningful representations and translational validation.
- **Multimodal and privacy-aware sensing:** radar, RGB, depth, skeleton and multimodal learning for sign language recognition and accessible healthcare communication.
- **Adaptive and distributed learning:** personalized continual/incremental learning, self-supervision, federated learning and prompt-based adaptation.
- **Efficient and embedded AI:** knowledge distillation, post-training quantization, edge inference, STM32/embedded deployment and resource-constrained optimization.
- **Applied machine intelligence:** cognitive assessment, industrial inspection, predictive reliability, anomaly detection, volcanology and data-driven engineering systems.

Research trajectory

2026

Efficient deep models

Knowledge distillation, post-training and mixed-precision quantization, sparsification and pruning for efficient vision models and deployment under compute constraints.

2025

Adaptive medical AI

Vision-language prompt tuning, personalized continual/incremental learning and federated learning for data-efficient medical image analysis.

2024

Digital cognitive assessment

Computer vision for cognitive screening, including facial micro-movement analysis and eye-gaze modeling for automated neurocognitive assessment.

2023–2026

Italian Sign Language, multimodal communication & privacy-aware sensing

Multimodal LIS recognition for healthcare communication using RGB, depth, skeleton and radar sensing; MultiMedaLIS, radar-based recognition and SignEval benchmarks/challenges.

2022–2023

Cardiovascular imaging & computational physiology

Coronary angiography, stenosis assessment and non-invasive FFR/iFR estimation, establishing the methodological foundations of the STARFLOW research line.

2020–2021

Volcanic seismology & anomaly detection

Deep learning for seismic anomaly detection and volcanic-unrest analysis with INGV/EUROVOLC, including the VolUnD toolkit and machine-learning database activities.

2020

Thoracic imaging & lung analysis

Deep-learning and computer-vision methods for lung and thoracic imaging, extending early medical-imaging work beyond neuroimaging.

2019–2020

Neuroimaging & brain dynamics

Early research on brain data and representation learning across EEG and fMRI, with emphasis on spatio-temporal patterns and neuroimaging analysis.

Patents, inventions and technology transfer

2026

Angiographic image processing method and system

Italian patent no. 102024000014434; PCT/IB2025/056252; WO 2025/262634 A1

Italian patent granted by UIBM on 20 July 2026. The patent family protects AI-based angiographic processing underpinning the STARFLOW line of research.

pending

Biometric safety system and method for estimating a user hazard condition and triggering an emergency stop

Italian patent application — under review

Inventors include Raffaele Mineo, Paolo Soda, Simone Palazzo and Concetto Spampinato.

in development

AI Framework for Automotive Microcontrollers

Patent filing / invention under development

Inventors include Francesco Rundo, Raffaele Mineo, Simone Palazzo and Concetto Spampinato.

Academic service and leadership

Editorial roles

- Associate Editor — Frontiers in Communication (2026–)
- Lead Guest Editor — Cognitive Computation, Special Issue “Cognitive and Multimodal Sign Language Processing: Recognition, Understanding, Translation, Generation, and Inclusive Interaction” (2026)

Area-chair, program and organizing roles

- Area Chair — MICCAI 2026
- Area Chair — IJCNN 2027
- Organizing Committee — IEEE ISBI 2026, Associate to Technical Program Chairs
- Paper Selection Committee — IPMI 2027

Workshops, challenges and special sessions

- Organizer — IEEE ISBI 2026 Special Session: “Privacy-Aware, Data-Efficient AI via Personalized Incremental and Federated Learning in Healthcare”
- Organizer — ICCV 2025: 1st Workshop on Multimodal Sign Language Recognition (MSLR 2025) + SignEval 2025 Challenge
- Organizer — ACM Multimedia 2025: (PILM '25) International Workshop on Personalized Incremental Learning in Medicine
- Organizer — CVPR 2026: 2nd Workshop on Multimodal Sign Language Recognition (MSLR) + SignEval 2026 Challenge

Journal reviewing

- IEEE Transactions on Medical Imaging (TMI), since 2025
- Pattern Recognition (Elsevier)
- Nature Communications, since 2025
- IEEE Open Journal of the Computer Society (OJCS), since 2025
- Computer Vision and Image Understanding (CVIU), since 2023
- Machine Vision and Applications (MVAP), since 2023
- ACM Transactions on Intelligent Systems and Technology (TIST), since 2024
- BioData Mining, since 2026
- Disability and Rehabilitation: Assistive Technology, since 2026
- Frontiers in Digital Health, since 2026
- International Journal of Cognitive Computing in Engineering (IJCCE), since 2026
- Journal of Imaging Informatics in Medicine (JIIM), since 2026
- IEEE Transactions on Biomedical Engineering (TBME), since 2026

Conference and workshop reviewing

- **Conferences:** ECCV 2024; BIBM 2024; RTSI 2024; ISBI 2026; MICCAI 2026; IJCNN 2026; EMBC 2026; NeurIPS 2026; AAAI 2027.
- **Workshops:** MICCAI AIPAD 2024; IJCNN SPAN 2025; CVPR MSLR 2026; ICPR CVBMC 2026; MICCAI AFRICAI 2026.

Volunteering, mentoring and research advising

- Research Advisor — RISE-MICCAI Publication Accelerator (2027)
- Mentor — Open Hackathons
- Mentor — GreenMindAI Catania Hackathon 2025
- Proctor — IEEEExtreme 20.0 (2026)

- Proctor Ambassador — IEEEExtreme 20.0 (2026)
- Student Volunteer — ACM Multimedia 2025

Awards, grants and recognition

- 2025** **ICCV 2025 Broadening Participation (BP) Travel Support**
IEEE/CVF International Conference on Computer Vision (ICCV 2025)
- Travel support awarded through the ICCV 2025 Broadening Participation program.
- 2025** **ACM Multimedia 2025 Student Travel Award**
33rd ACM International Conference on Multimedia (ACM Multimedia 2025)
- Student travel award from ACM SIGMM / the ACM Multimedia 2025 organizing committee.

Peer-reviewed journal articles

- 2024** **Non-invasive physiological assessment of intermediate coronary stenoses from plain angiography through artificial intelligence: the STARFLOW system ***
 Ovidio De Filippo*, **Raffaele Mineo***, Michele Millesimo, Wojciech Wańha, Federica Proietto Salanitri, Antonio Greco, Antonio Maria Leone, Luca Franchin, Simone Palazzo, Giorgio Quadri, Domenico Tuttolomondo, Enrico Fabris, Gianluca Campo, Alessandra Truffa Giachet, Francesco Bruno, Mario Iannaccone, Giacomo Boccuzzi, Nicola Gaibazzi, Ferdinando Varbella, Wojciech Wojakowski, Michele Maremmanni, Guglielmo Gallone, Gianfranco Sinagra, Davide Capodanno, Giuseppe Musumeci, Paolo Boretto, Pawel Pawlus, Andrea Saglietto, Francesco Burzotta, Marco Aldinucci, Daniela Giordano, Gaetano Maria De Ferrari, Concetto Spampinato, Fabrizio D'Ascenzo
European Heart Journal-Quality of Care and Clinical Outcomes
- 2024** **A Convolutional-Transformer Model for FFR and iFR Assessment From Coronary Angiography**
Raffaele Mineo, F Proietto Salanitri, G Bellitto, I Kavasidis, O De Filippo, M Millesimo, GM De Ferrari, M Aldinucci, D Giordano, S Palazzo, F D'Ascenzo, C Spampinato
IEEE Transactions on Medical Imaging

Conference and workshop publications

- 2026** **SignEval 2026 Challenges Results ***
 Ahmed Abul Hasanaath, **Raffaele Mineo**, Hamzah Luqman, Sarah Alyami, Maad Alowaifeer, Amelia Sorrenti, Gaia Caligiore, Sabina Fontana, Egidio Ragonese, Giovanni Bellitto, Federica Proietto Salanitri, Concetto Spampinato, Motaz Alfarraj, Mufti Mahmud, Simone Palazzo, Nour Imane Zeghib
IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops
- 2026** **Advancing Healthcare with Multimodal AI: Enhanced Diagnosis, Cognitive Monitoring, and Inclusive Medical Communication**
Raffaele Mineo
International Conference on Medical Image Computing and Computer-Assisted Intervention
- 2026** **A Benchmark for Radar-Based Italian Sign Language Recognition using Frequency-Domain Range-Time Maps**
Raffaele Mineo, Amelia Sorrenti, Gaia Caligiore, Federica Proietto Salanitri, Giovanni Bellitto, Senya Polikovsky, Sabina Fontana, Egidio Ragonese, Concetto Spampinato, Simone Palazzo
IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops
- 2025** **Timely Predicting Power Mosfet Failure in EV: AI Based Metodology and Deployment at the Edge**
 Robin Faro, Francesco Cancelliere, Alessssandro Strano, **Raffaele Mineo**
2025 IEEE Symposia on Computational Intelligence for Energy, Transport and Environmental Sustainability (CIETES Companion)
- 2025** **The SignEval 2025 Challenge at the ICCV Multimodal Sign Language Recognition Workshop: Results and Discussion ***
 Hamzah Luqman*, **Raffaele Mineo***, Murtadha Aljubran, Ahmed Abul Hasanaath, Amelia Sorrenti, Sarah Alyami, Sadam Al-Azani, Maad Alowaifeer, JiHwan Moon, Vaclav Javorek, Tomas Zelezny, Marek Hruz, Gaia Caligiore, Silvio Giancola, Senya Polikovsky, Motaz Alfarraj, Sabina Fontana, Mufti Mahmud, Muhammad Haris Khan, Kamrul Islam, Sevgi Gurbuz, Egidio Ragonese, Giovanni Bellitto, Federica Proietto Salanitri, Concetto Spampinato, Simone Palazzo
Proceedings of the IEEE/CVF International Conference on Computer Vision

- 2025 Text-Aligned Radar-Based Sign Language Recognition for Healthcare Communication**
Raffaele Mineo, Amelia Sorrenti, Gaia Caligiore, Federica Proietto Salanitri, Giovanni Bellitto, Senya Polikovsky, Sabina Fontana, Egidio Ragonese, Concetto Spampinato, Simone Palazzo
Proceedings of the IEEE/CVF International Conference on Computer Vision
- 2025 Radar-Based Imaging for Sign Language Recognition in Medical Communication**
Raffaele Mineo, Gaia Caligiore, Federica Proietto Salanitri, Isaak Kavasidis, Senya Polikovsky, Sabina Fontana, Egidio Ragonese, Concetto Spampinato, Simone Palazzo
International Conference on Medical Image Computing and Computer-Assisted Intervention
- 2025 PCB-SAID: A Low-Cost Camera-Based Dataset for Few-Shot SMD Assembly Inspection**
Raffaele Mineo, Amelia Sorrenti, Robin Faro, Gabriele Mineo, Francesco Cancelliere, Alberto Faro
Proceedings of the IEEE/CVF International Conference on Computer Vision
- 2025 Memory-Augmented Prompt Tuning for Self-Supervised Continual Learning in Medical Imaging**
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International Workshop on Personalized Incremental Learning in Medicine (PILM)
- 2025 Learning Joint Text and Visual Tokens in CLIP for Medical Image Analysis**
Raffaele Mineo, Giovanni Bellitto, Federica Proietto Salanitri, Rutger Hendrix, Concetto Spampinato, Simone Palazzo
Proceedings of the 2nd International Workshop on Multimedia Computing for Health and Medicine
- 2025 HC-PTQ: Poincaré-Based Hyperbolic Clustering for Data-Free Quantization of Vision Transformers**
Raffaele Mineo, Simone Palazzo, Concetto Spampinato, Francesco Rundo
Proceedings of the IEEE/CVF International Conference on Computer Vision
- 2025 Conditional Visuo-Textual Prompt Learning for Medical Image Analysis**
Raffaele Mineo, Giovanni Bellitto, Federica Proietto Salanitri, Rutger Hendrix, Ulas Bagci, Daniela Giordano, Concetto Spampinato, Simone Palazzo
2025 IEEE 22nd International Symposium on Biomedical Imaging (ISBI)
- 2025 Automated MoCA Score Estimation Using Eye-Gaze Data and Vision Transformers**
Raffaele Mineo, Isaak Kavasidis, Federica Proietto Salanitri, Lisa Passarello, Giovanni Piccininno, Vincenzo Masciale, Alessandro Anselmo, Cristiano Convertino, Domenico Rotondi, Nicola Laurieri, Simone Palazzo, Concetto Spampinato, Manuela Pennisi, Daniela Giordano
2025 IEEE 38th International Symposium on Computer-Based Medical Systems (CBMS)
- 2024 Sign Language Recognition for Patient-Doctor Communication: A Multimedia/Multimodal Dataset**
Raffaele Mineo, Gaia Caligiore, Concetto Spampinato, Sabina Fontana, Simone Palazzo, Egidio Ragonese
2024 IEEE 8th Forum on Research and Technologies for Society and Industry Innovation (RTSI)
- 2024 Multisource Approaches to Italian Sign Language (LIS) Recognition: Insights from the Multi-MedaLIS Dataset ***
 Gaia Caligiore*, **Raffaele Mineo***, Concetto Spampinato, Egidio Ragonese, Simone Palazzo, Sabina Fontana
2024 10th Italian Conference on Computational Linguistics (CLiC-it)
- 2024 Magnifying facial micro-movements for cognitive evaluation**
Raffaele Mineo, Federica Proietto Salanitri, Lisa Passarello, Alberto Sardella, Laura Messina, Manuela Pennisi, Daniela Giordano, Simone Palazzo, Concetto Spampinato
2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)
- 2024 Coronary Artery Stenosis Assessment in X-Ray Angiography Through Spatio-Temporal Attention for Non-Invasive FFR and iFR Estimation**
Raffaele Mineo, Federica Proietto Salanitri, Giovanni Bellitto, Ovidio De Filippo, Fabrizio D'Ascenzo, Simone Palazzo, Concetto Spampinato
Proceedings of the 17th International Joint Conference on Biomedical Engineering Systems and Technologies - Volume 1: BIOIMAGING
- 2023 FeDETR: A Federated Approach for Stenosis Detection in Coronary Angiography**
Raffaele Mineo, Amelia Sorrenti, Federica Proietto Salanitri
International Conference on Image Analysis and Processing
- 2023 Ensemble and personalized transformer models for subject identification and relapse detection in e-prevention challenge**
 Salvatore Calcagno, **Raffaele Mineo**, Daniela Giordano, Concetto Spampinato
ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)
- 2023 Dynamic Graph Attention: Unraveling Spatio-Temporal Synchrony in EEG Data**
 Federica Proietto Salanitri, Giovanni Bellitto, **Raffaele Mineo**, Matteo Pennisi, Amelia Sorrenti, Salvatore Calcagno, Daniela Giordano, Simone Palazzo, Concetto Spampinato
2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)
- 2023 BioTrak: A Blockchain-based Platform for Food Chain Logistics Traceability**
 Alessia Spitaleri, Isaak Kavasidis, Vincenzo Cartelli, **Raffaele Mineo**, Francesco Rundo, Simone Palazzo, Concetto Spampinato, Daniela Giordano
2023 International Conference on Intelligent Computing, Communication, Networking and Services (ICCNS)

Other scholarly and research outputs

- 2025** **Proceedings of the International Workshop on Personalized Incremental Learning in Medicine (PILM '25)**
Simone Palazzo, Giovanni Bellitto, **Raffaele Mineo**, Amelia Sorrenti, Camila Gonzalez, Angelo Porrello, Lorenzo Bonicelli
ACM Multimedia 2025
- 2021** **VOLcano UNrest Detection (VolUnD) Toolkit**
R. Mineo, P. Spadaro, F. Cannavò, S. Palazzo, A. Cannata, C. Spampinato
Zenodo
- 2021** **EUROVOLC - Deliverable Report - D9.3 Machine learning tool kit and database**
S. Palazzo, **R. Mineo**, C. Spampinato, F. Cannavò, A. Cannata, C. Montagna, J. M. Saurel, O. Geber, A. Lemarchand
HORIZON 2020 EUROVOLC Project

Current / additional scholarly outputs represented in the supplied metadata

- n.d.** **Personalized Incremental Learning in Medicine: A Comprehensive Survey of Methods, Challenges, and Future Directions**
-
Manuscript / scholarly output
- n.d.** **Anomaly Detection in Volcanic Seismic Data Using Deep Learning: A Case Study on Mt. Etna**
-
IEEE Access — metadata entry; status/year to update

Research projects and collaborations

- **HORIZON Europe — NEUROKIT2E** (Grant Agreement 101112268): research on knowledge distillation and meta-learning for on-chip AI inference; contributions also spanning embedded/edge AI and performance evaluation.
- **HORIZON Europe — EdgeAI related activities**: deep-learning algorithms for defect detection on electronic boards and wood products; multi-GPU and edge-computing experimentation; neuromorphic approaches; KPI and functional/non-functional requirement evaluation.
- **HORIZON 2020 — EUROVOLC** (Grant Agreement 731070): machine-learning toolkit and database for volcanic-unrest analysis; VolUnD toolkit and D9.3 deliverable.
- Research collaborations with the **Istituto Nazionale di Geofisica e Vulcanologia (INGV)**, Catania; **AOU Città della Salute e della Scienza**, Turin; **Max Planck Institute for Intelligent Systems**, Tübingen; and **STMicroelectronics R&D Power and Discrete**, Catania.
- Industrial R&D / consulting collaboration with **DeepSensing S.r.l.**, Catania.

Teaching

- Teaching Tutor, **Database and Web Programming**, Computer Engineering, University of Catania — multiple editions, 2022–2025.
- Teaching Tutor, **Fundamentals of Computer Science**, Industrial Engineering, University of Catania — 2022–2023.

Professional memberships and technical communities

- CVPL / PeRCeiVe research community (since 2019)
- IEEE Engineering in Medicine and Biology Society (EMBS), since 2024
- IEEE Signal Processing Society (SPS), since 2025
- Association for Computing Machinery (ACM), since 2025
- ACM SIGMM, since 2025
- IEEE Computer Society Technical Community on Pattern Analysis and Machine Intelligence (TCPAMI), since 2026

- IEEE Engineering in Medicine and Biology Society (EMBS) Technical Committee on Biomedical Imaging and Image Processing (BIIP), since 2026

Technical competencies

- **Artificial intelligence and data:** AI for Medicine, machine learning, deep learning, data acquisition systems, PyTorch, Keras, TensorFlow, R, SQL, MySQL, PostgreSQL.
- **Programming:** Python, C, C++, C#, Java, JavaScript, TypeScript, Go, PHP, AutoIt, OpenGL, QML, Qt Framework / Qt Quick, ASP.NET, Angular, Laravel, Node.js.
- **Embedded systems and sensing:** STM32 firmware, Arduino, Infineon Radar SDK, Intel RealSense SDK, Microsoft Kinect SDK, Stereolabs ZED SDK, edge computing, multi-GPU systems.
- **Engineering and development tools:** Android Studio, Atollic TrueSTUDIO, Eclipse, IntelliJ IDEA, Qt Creator, Visual Studio, Visual Studio Code, AutoCAD, Autodesk EAGLE, Autodesk Tinkercad.
- **Web, systems and networks:** HTML, CSS, SCSS, JSON, client/server applications, network architecture, operating systems.
- **Media, graphics and productivity:** Adobe Illustrator, Photoshop, Acrobat, After Effects, Premiere Pro, Blender, CorelDRAW, Unity, Microsoft Office, Excel and PowerPoint.

Languages

- Italian — native
- English — B1 (Cambridge certified)
- French — A1
- Spanish — A1

Professional qualifications, certifications and training

- Italian State Examination and professional qualification as Information Engineer (2021)
- Mensa Italia certificate / membership credential
- ECDL Core AICA (2014)
- Cambridge English B1 Preliminary (2014); Cambridge English A2 Key (2012)
- Würth Elektronik Italia — Power Expert technical training (2021)
- Coursera / DeepLearning.AI: Neural Networks and Deep Learning; Improving Deep Neural Networks; Structuring Machine Learning Projects; Convolutional Neural Networks; Sequence Models; Deep Learning Specialization
- Coursera: AI for Medical Diagnosis; AI for Medical Prognosis; AI for Medical Treatment; AI for Medicine Specialization; Detecting COVID-19 with Chest X-Ray using PyTorch
- Coursera / DeepLearning.AI: Build Basic GANs; Build Better GANs; Apply GANs; Generative Adversarial Networks Specialization

Additional hands-on technical background

Electronic-board repair for household appliances; desktop/laptop/smartphone/tablet hardware and software repair; installation and maintenance experience with television systems, alarm systems, video surveillance, structured cabling and selected plumbing/HVAC systems; electro-welding and carpentry; preparation of technical-estimative reports and IT appraisals.